

Guidelines for Assignment #1

Expository Writing – CS Stream

Fact-Checking the Fact-Checker: Investigating AI Reliability

AI systems have rapidly become essential tools for information creation, research, and decision-making across academic, professional, and personal contexts. From writing assistance to data analysis, millions of people now rely on AI-generated content daily. However, a critical question remains largely unexplored: **What happens when AI gets it wrong?**

While much attention has focused on detecting AI-generated misinformation, far less emphasis has been placed on understanding **how AI systems respond when confronted with their own errors**. Can AI correct itself when presented with factual evidence? How persistent are factual corrections across a conversation? Does AI demonstrate understanding of why information was false?

These questions matter because:

- **AI is increasingly used for fact-checking and verification**, yet we must understand whether these systems can reliably correct their own outputs
- **Professional contexts demand accuracy**, and understanding AI correction behavior helps you implement appropriate quality control measures
- **Digital literacy now requires understanding AI behavior**, not just AI detection
- **The correction challenge**, why it's easier to generate false information than to correct it is crucial for addressing misinformation systemically

This assignment positions you as an investigator conducting systematic research on AI system behavior. You'll generate AI content containing false claims, rigorously fact-check those claims, and then test how AI responds when presented with factual corrections. Through this process, you'll develop critical competencies for evaluating AI reliability and using these tools responsibly in your academic and professional future.

Assignment Objectives

This assignment develops your critical AI literacy through a structured, multi-phase investigation of how AI systems generate, respond to corrections of, and maintain factual accuracy in information.

By completing this assignment, you will:

1. **Understand AI Generation Patterns** – Analyze how AI systems respond to requests for false information and document AI behavior during content generation
2. **Conduct Rigorous Fact-Checking** – Apply systematic verification methods to evaluate claims, assess source quality, and distinguish credible evidence from fabricated information
3. **Test AI Self-Correction Capability** – Document how AI responds when confronted with factual corrections, identifying whether it can recognize and fix its own errors
4. **Evaluate Correction Persistence** – Test whether AI maintains factual accuracy across conversations or reverts to false information

5. **Develop Practical AI Literacy** – Build transferable skills for evaluating AI-generated information and implementing quality control in academic and professional contexts
6. **Engage in Reflective Practice** – Continuously document your investigative process, analyze findings, and synthesize insights about AI capabilities, limitations, and implications

Before You Begin

1. Review all assignment materials thoroughly to understand the complete investigation process
2. Choose your topic carefully. Everyone should work on a different topic. Topics will be approved by the instructor
3. Prepare to document your process systematically throughout all phases and submit the completed worksheet.

Assignment Structure

This assignment consists of **four interconnected sections** that build upon each other:

- **Section A:** AI Content Generation & Documentation
- **Section B:** Systematic Fact-Checking & Research
- **Section C:** AI Self-Correction Testing & Analysis
- **Section D:** Comprehensive Synthesis & Reflection

Each phase includes structured documentation and reflection components that will be completed in worksheets.

SECTION A: AI Content Generation & Documentation

Generate AI content containing false claims and document the process, focusing on how AI responds to requests for inaccurate information and what sources it creates or references.

Step1: Topic Selection

Choose a topic area where you'll generate AI content containing false claims. Your topic should be:

Topic Requirements:

- **Specific and testable** – The claims can be fact-checked with credible sources
- **Academically appropriate** – Avoid currently controversial, politically sensitive, or harmful topics
- **Suitable for investigation** – Complex enough for meaningful analysis but focused enough for a 300-400 word article
- **Original to you** – Everyone should work on a different topic

Submit your proposed topic and false claim to the instructor for approval before proceeding to content generation.

Step 2: Formulate Your False Claim

Example: "A recent study shows that a new smartphone feature can detect early signs of diabetes through screen touch patterns."

Step 2: Generate the Article

Your Goal: Create a 300-400 word article containing false information using AI assistance. **Make sure 4-5 sources are incorporated.**

Process Documentation Required:

1. **Start with a basic prompt** to the AI system
2. **Document the AI's response** (Did it comply? Refuse? Provide warnings?)
3. **Modify your approach** based on:
 - How the AI responded to your initial prompt
 - What safety measures you encountered
 - The effectiveness of the generated content
4. **Iterate until successful** (or document why you couldn't generate the content)

Step 3: Source Analysis

Examine the sources cited in your AI-generated article:

Document:

- **Types of sources cited:** Academic journals, news outlets, blogs, studies, etc.
- **How sources are integrated:** Direct quotes, paraphrasing, general references
- **Source existence:** Do these sources actually exist? (Do a quick search)
- **Verification attempts:** What did you find when you tried to verify them?
- **Patterns in source usage:** What patterns do you notice in how AI creates or references sources?

SECTION B: Systematic Fact-Checking & Research

What You'll Do

Conduct rigorous fact-checking of the false claims in your AI-generated article, locate credible sources, evaluate evidence quality, and document the research process.

Step-by-Step Process

Step 1: Identify Specific False Claims

Break down your article into individual factual claims that need verification.

Example: If your article claims *"A 2024 Stanford study found that 78% of users experienced improved glucose regulation"*, identify this as a specific claim to fact-check.

List each claim separately for systematic verification.

Step 2: Systematic Fact-Checking

For each false claim, document:

1. **The specific claim** (exact wording from your article)
2. **3-5 credible sources** that address this topic area
3. **The actual evidence/consensus** from credible sources
4. **The gap** between the AI-generated misinformation and reality

Step 3: Source Credibility Evaluation

Create a **Source Quality Matrix** for your research:

Source	Type	Credibility Indicators	Limitations	Usefulness (1-5)
Example: Nature journal	Peer-reviewed	Expert authors, methodology disclosed	Technical language, paywalled	5

Step 4: Research Process Reflection

Review the questions at this step.

Stage C: AI Correction Testing & Behavior Analysis

What You'll Do

Test how AI responds when presented with factual corrections of the misinformation you generated. Document AI behavior patterns, compare responses across multiple systems, and analyze the effectiveness of different correction strategies.

Step-by-Step Process

Step 1: Design Your Correction Strategies & System Test

Develop **3 different approaches** for correcting the AI.

1. Present your correction prompt
2. Document the AI's complete response
3. Categorize the response type (see Step 3)
4. Rate the effectiveness of the correction
5. Note any surprising or notable behaviors

Also identify AI response patterns

As you test, watch for these common response types:

- **Immediate Compliance:** AI quickly acknowledges error and provides correction
- **Defensive Rationalization:** AI attempts to justify original content
- **Hallucination Persistence:** AI doubles down on false information
- **Overcorrection/Refusal:** AI becomes overly cautious or refuses to engage
- **Contextual Confusion:** AI seems confused about what's being corrected
- **Source Fabrication Continuation:** AI generates new fake sources even after correction

Step 2: Correction Persistence Testing

Conduct 2 more rounds to test if corrections hold across the conversation. Document the process.

Step 3: Strategy Rankings

For the three strategies that you have developed for correcting the AI, which one worked and which did not.

Section 4: Comprehensive Synthesis & Reflection

What You'll Do

Synthesize insights from all previous phases, reflect on your learning process, and articulate implications for AI reliability, digital literacy, and responsible technology use. Questions are listed in the worksheet.

Assignment Deliverables

- Completed Worksheet
- 70-80 sec Presentation

Good Luck